

Study of Linguistic Features Incorporated in a Literary Book Recommender System

Haifa Alharthi
University of Ottawa
Ottawa, Canada
halha060@uottawa.ca

Diana Inkpen
University of Ottawa
Ottawa, Canada
Diana.Inkpen@uottawa.ca

ABSTRACT

As reading for pleasure is declining, particularly among the young, we are motivated to investigate the textual elements that may play a role in generating high-quality book recommendations. Two recommendation algorithms were trained on more than a hundred features learned from the full text of the books. Both algorithms generate more accurate lists of suggested books than many competitive recommender systems. Exploration of the key textual elements is conducted together with qualitative analysis.

CCS CONCEPTS

• Information systems → Recommender systems; • Computing methodologies → Natural language processing;

KEYWORDS

Recommender Systems, Book Recommendations, Stylometry features

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1 INTRODUCTION

Due to the vast number of products available online, internet users can have difficulty finding merchandise that meets their needs. Recommender systems (RS) address this by suggesting items that the target user might be interested in. This paper proposes to use linguistic features (including stylometry) in a book recommender system that would be useful for libraries and educational institutions. Book RSs can help readers choose from the enormous number of books available: more than 129,864,880 titles as reported by Google in 2010¹.

Though e-books provide readers with an abundance of easy-to-access resources, reading for pleasure is declining, particularly

among the young.² This is unfortunate since numerous studies suggest reading has many benefits. A significant difference was found when the well-being of 7,500 Canadian readers was compared to non-readers in [19]. The first group reportedly had better physical and mental health, were more inclined to volunteer and had comparatively higher satisfaction. Fiction, in particular, has been found to encourage social communication [26], correlate with greater empathy and social skills [27] and has prolonged biological impact, especially in brain connectivity [5]. Therefore, deploying machine learning algorithms to promote reading for pleasure by automatically predicting the most interesting kind of books for individuals is a boon for readers.

There are two main types of recommender systems: collaborative filtering (CF) and content-based recommendation (CB). CF requires a rating matrix of all other users in the system to predict a target user's reading preferences, and since the matrix can be large and sparse it could result in inaccurate recommendations. Sparsity issues are common in libraries where many books have never been checked out (e.g. 75% of the books in the library of the Changsha University of Science and Technology [47]). Sparsity results in the problem of user/item cold start which make recommendations difficult, as there may be too few ratings. Other CF problems include *the long tail* issue (also called *the Harry Potter issue*) which occurs when a few popular items divert attention from the majority of items, the *shilling attack* which is caused by fake ratings (i.e. promotion) and the *gray sheep* who are users with completely different tastes than others [1].

Content-based recommender systems analyze the content of the consumed items, such as book genres and authors, to predict future preferences. CB does not have issues related to rating matrix since it relies solely on the target user ratings. Item cold start is not an issue in CB, as items without any ratings can be recommended immediately, if they are similar to the user previous readings. Moreover, CB recommendations are easy to interpret; for example, books can be recommended if they share a genre. However, CB cannot accurately suggest items without adequate content, and it is also prone to *overspecialization*, generating non-diverse recommendations [1]. Many problems with both systems can be avoided by using a hybrid RS.

This paper proposes a content-based recommender system that considers books' textual features. A recent survey on book RSs [1] showed that few systems have analyzed actual book text. Though literary analysis is an active area (e.g., narrative structure recognition [15]) such examination is not common with book RSs. Unlike other text-based items (e.g., news, scientific articles), many book-specific factors affect reading preferences. Reader's advisory in libraries, a

¹<http://tinyurl.com/24vypnk>

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²<http://tinyurl.com/jgunwfx>

service to offer patrons suggestions, considers seven appeal factors: *characterization frame, language and writing style, pacing, special topics, storyline, and tone*. One can pay to access information about appeal factors using reader-advisory databases such as NoveList³, which was created by professionals [32].

Analysis of book text to suggest recommendations is already adopted by many commercial organizations, including BookLamp⁴. Google offers a search engine that retrieves books by using the actual text, and E-book readers such as Kindle also provide users with book promotions. Hence, the implementation of our content-based system may help current e-book recommenders, and improve online stores.

The contributions of this paper are as follows. We employed in a book RS multiple linguistic and stylistic aspects (120) learned from book text including lexical, syntactic and fiction-specific features (e.g., the number of fictional characters).⁵ We are not aware of any work that includes and analyzes all these types of features for book recommendations. Our system provides significantly better recommendations than state-of-the-art CB systems such as Vector space model (VSM), latent semantic indexing (LSI), latent Dirichlet allocation (LDA), Doc2vec and a CF system, namely matrix factorization (MF). We also compared with another book RS [2] which takes advantage of books' texts. One of the two adopted recommendation methodologies ranks books using a forest of randomized regression trees, and its variables importance can provide insight into the factors that contributed to the accuracy of the recommendations. The identification of elements that contributed the most to users' book preferences may help the system owners and publishers to understand factors that appeal to individuals and communities. We aggregate and analyze the most and the least significant variables for each user; then, we conduct further qualitative analysis by studying the genre and the authors of similar books.

2 RELATED WORK

Current book RSs adopt traditional methods such as CF [42][44], analyze library loan records [40] and assess user and communities' textual reviews [32–34]; only a few systems take the actual text of books into account. A CB system in [41] is based on the limited stylometry features of document length, vocabulary richness, part-of-speech bigrams, and most frequent words. The system considers two methods of book representation: LDA topics and the stylometric features. The recommendations are generated by the Rocchio algorithm [25] and evaluation by LitRec dataset [43] resulted in a hybrid system with superior performance when combining CB and CF, compared to individual CF or CB systems. Comparing the book representations, the LDA-based approach gave the best recommendation accuracy. In our proposed system, only two features out of 120 are similar to [41] namely document length and vocabulary richness.

Book text was also exploited in [48] which recommends authors and books. Items are represented in a hierarchical structure of four layers, using a multilayer self-organizing map (MLSOM) [36]. Author information is in the first layer, and the other three layers

are the author's books, pages of every book and paragraphs on each page. The quality of the system is evaluated by analyzing the relevance of two books; that is, where a relevant recommendation is a book that shares the same genre. Compared to VSM, LSI and other systems, this approach has higher accuracy.

Another book RS was proposed by [2], and represents books as the penultimate layer in convolutional neural networks that learns from books' sequence of words or character bigrams to predict their authors. The book representations (named AuthID) are then fed into a regressor (support vector regression) which finds pattern of books read by the user. This method reportedly outperforms state-of-the-art systems, including VSM, LDA and LSI. In our paper, we also compare the use of our linguistic features against AuthID representations (see section 5).

The baselines are commonly adopted in text-based recommender systems. A vector space model (VSM), the basic information retrieval approach, was used to find similar books based on their descriptions in [33, 40] and not their actual text. Other popular information retrieval approaches using topic-modeling techniques, namely latent semantic indexing (LSI) and latent Dirichlet allocation (LDA) are also used in textual item recommendations such as [4] to recommend movies based on plots and [23] and [28] to recommend articles. A more recent state-of-the-art approach is paragraph2vec, also called Doc2vec, which was proposed by [22] to provide a fixed-size representation for paragraphs and documents. It works in an unsupervised fashion to predict the words in texts of various lengths (either a paragraph or a large document) and produce an embedding for each document/paragraph. In recommender systems, Doc2vec was used in [18] and [45] to recommend scientific articles and answers in question-answering systems respectively.

3 METHODOLOGY

3.1 Features selection

We considered multiple feature sets learned from book texts, with the exception of the author's name, converted into a numerical value using one-hot encoding. We exploit book texts directly instead of relying on labels tagged by experts e.g., genre, for the following reasons. The list of available labeled books is far from complete due to the difficulty of manually processing the high number of books. For example, in some of our experiments, we did not find information about specific authors/books in NoveList. Furthermore, the assignment of genres can adopt a subjective scheme [37].

Linguistic Inquiry and Word Count (LIWC) is a popular resource that was originally developed for psychological analysis, and thus focuses on psychological in addition to grammatical and content word categories. We computed all categories using the LIWC 2015 dictionary, which contains more than 6,400 terms including word stems and emoticons. Each of the 92 categories ([30] explains each feature) has associated sub-dictionary. Whereas many LIWC classes simply reflect their names (e.g., articles dictionary has only three words "a", "an" and "the."), other classes are constructed by experts (e.g., the emotion category). An agreement of two out of three judges is needed to add or remove a word from a dictionary. A word may belong to multiple dictionaries such as sadness and emotions. The value of a category is the percentage of its words in the book (i.e., 4.54% of words in a book are auxiliary verbs) [39].

³<https://www.ebscohost.com/novelist>

⁴<http://www.businessinsider.com/apple-buys-booklamp-2014-7>

⁵To download: <https://tinyurl.com/y9svvgwg>

Table 1: Categories of features included in our study

Category	Feature Names
Lexical	<i>Token-based</i> : average length of paragraph, sentence and words; average number of commas per sentence; average variance in paragraph length and in sentence length; length of book; words>6-letters; dictionary words; average syllables per word Lexical density (type-token ratio) <i>Word frequencies</i> : percent of latinate words; function words; affect words; social words; cognitive processes; perpetual processes; biological processes; core drives and needs; time orientation; relativity; personal concerns ⁶
Character-based	Numbers; all punctuations
Syntactic	Percentage of adjectives, adverbs, nouns, and verbs; comparatives, interrogatives; quantifiers
Characterization	Percentage of dialog text; number of dialogs; average length of dialog texts; number of unique fictional characters; number of times fictional characters are mentioned; number of unique places; number of times places are mentioned; Percentage of dialogue by female characters; Percentage of characters which are female
Six-styles	Literary; abstract; objective; colloquial; concrete; subjective

Most considered Linguistic features have been applied broadly in literature analysis; however, not for book recommendations. LIWC was used for analyzing the profiling of authors [13] and fictional characters' gender [14]. One quantitative analysis of genre profiling is conducted in [29] which tests seven directional hypotheses about the differences in content and style of science fiction, mystery and fantasy. Measurements of content and style of carefully-selected literary books are conducted using seven LIWC categories. It is found that compared to mystery, science fiction has more cognition and religious terms and less social words, pronouns and auxiliary verbs. Compared to fantasy, science fiction has a higher rate of cognition word and auxiliary verbs*⁷, and a lower rate of social, biological* and religious terms*. Also, fantasy has significantly more biological* and religious* term than mystery. These previous studies encourage us to apply content and style word classes to make book recommendations.

Other text measurements are computed by using GutenTag [9] which, similarly to our dataset, was built on the Project Gutenberg texts. GutenTag's built-in tagger uses a stylistic lexicon [11, 12] to calculate stylistic aspects usually considered when analyzing English literature. Text styles are not meant to exclusively reflect aesthetical characteristics, but to also represent generic differences of texts such as of genres. The six styles are colloquial (informal) vs. literary (more specific to literature), concrete (related to physical objects) vs. abstract (related to complex concepts) and subjective (influenced by feelings) vs. objective (not affected by emotions) [10, 11]. The six styles models helped in literature analysis; for example, it is found in [10] that free indirect discourse is quantitatively distinguishable from and centered between direct discourse and narration.

In addition, some elements specific to fictional works are also considered. As Bakhtin suggests [3], the number of fictional characters can provide information about the type of book. A small number of characters could indicate that the story takes a place in a rural setting where characters typically know one another

(e.g., family), while many characters suggest urban settings where interaction with unknown people (e.g., in a theatre) occur [15]. To identify characters, we used the fiction-aware named entity recognizer (LitNER)[8], which outperforms other popular POS taggers. We counted the number of characters in a book, and how many times they were mentioned. Since LitNER also recognizes locations, we counted the number of places in a novel as well. One appeal factor in the reader's advisory is *setting*, which is related to where and when the story takes place.

Moreover, similar to other book RSs [31–34] which were directed at emergent readers, we incorporate a text readability measurement. We calculated the Flesch reading ease score using textstat⁸. We also computed the average number of syllables per word by pyphen⁹.

All the previously mentioned measurements are grouped under stylometric categories in Table 1. Stylometric features are learned from text to detect the author, and according to [38] they consist of five categories: lexical, character-based, syntactic, semantic and application-specific. The application-specific features help identify an author's writing style in a specific domain, and characterization features could fall under this category.

3.2 Recommendation procedure

In content-based recommendations, for each user, a classifier or regressor is created to learn from previous preferences to predict future interests. In this section, we describe two approaches which empirically provided the best results.

K-nearest neighbors (KNN) is commonly used in RSs which, in our case, takes the input of books represented as vectors of features and find the most similar ones. A predicted score for book B_j , which is unrated by user u , is computed as the average of similarity values between B_j and K number of books rated by u . The similarity between two items is first calculated at the feature level, then averaged. Books with highest similarity values are recommended in the top k list. We experimented with multiple distance measures.

⁷* indicates statistical difference

⁸ <http://neon.niederlandistik.fu-berlin.de/en/textstat/>

⁹ <http://pyphen.org/>

The Gaussian kernel distance (for numerical features) [35] and the cosine similarity (1-cosine as the distance, for categorical features) gave the best results.

The other approach constructs for each user a forest of randomized regression trees. A single tree-based model [7] uses the training sample of size N , where each input X is a vector $(X_1, X_2, \dots, X_P)$ labeled with output Y , to build a tree T with internal and external nodes (leaves) representing features and labels respectively. The feature space χ is divided at the tree root into sub-regions which is recursively divided at each node t . At an internal node t , a test $s_t = (X_m < c)$ is performed to further split its sub-region into t_L and t_R ¹⁰. In regression trees, an external node t is labeled with $\bar{y}(t)$, which is the average output in t subset [24].

To ensure sub-regions with as homogeneous output as possible, the algorithm finds the best split which maximizes the impurity reduction $i(t)$ measured when splitting Nt (samples at node t) into two partitions t_L and t_R where $p_L = N_{tL}/N_t$ and $p_R = N_{tR}/N_t$. In equation 1, $i(t)$ refers to any impurity reduction such as variance in regression trees.

$$\Delta i(s, t) = i(t) - p_L i(t_L) - p_R i(t_R) \quad (1)$$

The growth of a tree is stopped once stopping criteria is met. After the model is constructed, a new sample can be propagated through the tree and eventually labeled with $\hat{y}$, the value on the leaf it ends up at [24].

Ensemble approaches, such as randomization-based algorithms, are adopted to enhance the performance of single trees that may have low accuracy due to high variance. Random Forests (RF) [6], a well-known extension of tree-based models, constructs trees based on multiple bootstrapped copies, and considers a random subset of a number of features at each node when deciding the best split. Another ensemble method is Extremely Randomized Trees, also called Extra Trees (ET), [16], which provides competitive results in a learning to rank task in [17]. ET generates multiple trees from the whole training set, with no bootstrapping required. When performing a split, ET also considers a random subset of features. ET takes the randomness a step further by choosing the split point at each internal node at random [17, 24].

In addition to their high accuracy, forests of randomized trees can provide insight regarding the reasons for a good performance by identifying key variables for predicting the output. One way to calculate the importance of a feature X_m is through Mean Decrease Impurity importance (MDI), as in equation 2. In the context of RF and ET, the average of MDI for all N_T trees is used [24].

$$\text{imp}(X_m) = \frac{1}{N_T} \sum_T \sum_{t \in T: v(s_t)=X_m} p(t) \Delta i(s_t, t) \quad (2)$$

$p(t) \Delta i(s_t, t)$ refers to the weighted impurity reduction for all internal nodes t where a feature X_m appears, $p(t)$ is $\frac{N_t}{N}$ and $v(s_t)$ is the relevant feature.

¹⁰In case of binary test.

4 EVALUATION

4.1 Dataset and Preprocessing

Experiments were conducted on a Litrec dataset [43], which has ratings of 1,927 users of 3,710 literary books. The dataset contains the complete Gutenberg text of books tagged with part-of-speech labels [42]. The user ratings are extracted from Goodreads¹¹, where 1 or 2 stars indicate dislike and 3 to 5 represents like; we adopted this range when converting the ratings to binary. All books rated with zero (17,976 ratings) are disregarded because in Goodreads that means the book was read but not rated. Similar to many CB related works, such as [46], we filtered out any user with less than 10 ratings to provide adequate data for the CB. To calculate the characterization features, any book without characters was excluded (Only 71 titles). The remaining number of users was 367, and they rated 1,050 unique books authored by 410 writers. The average length of book texts is 91580.07 tokens.

4.2 The Experimental Setting

The systems are evaluated in a top-k recommendation scenario, where each system provides a ranked list of the most relevant books to the target user. In content-based RSs, a user's books are divided into training and testing sets, and the CB learns the user preference only from the training set. Though the same split of training and testing sets per user is applied in the MF system, MF is trained on the whole rating-matrix excluding the target user's ratings on the items in the test set. For a user, each system generates a ranked-list from all the books except those in the user's training set. In an ideal system, books from the test set would be placed in the top ranks, since they are considered more relevant than the unread books. As with many related works, we assessed the recommendations at $k=10$. A three-fold cross-validation is performed for every user, and the statistical significance is measured by t-test at p-values of 0.05 or 0.01.

Metrics. The accuracy of the top-k recommendation is measured by precision at k and recall at k —Equations 3-4—where a *relevant* book is a preferred book, and it is *recommended* if it is ranked in the top k list.

$$\text{Precision}@k = \frac{\# \text{ relevant books in top } k \text{ recommended}}{k} \quad (3)$$

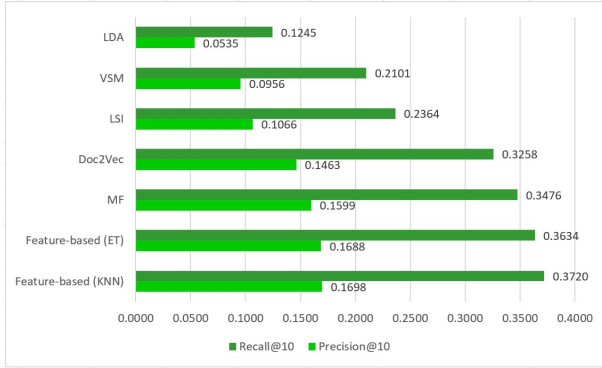
$$\text{Recall}@k = \frac{\# \text{ relevant books in top } k \text{ recommended}}{\# \text{ of relevant books}} \quad (4)$$

Baselines. We implemented LDA, LSI, VSM and Doc2vec using gensim¹², a Python library. The texts were tokenized and lower-cased, and NLTK stopwords and the least frequent words were removed. In LDA, LSI, and VSM, relevant books from a target user's training set are aggregated to create a query. In VSM, the recommended books are the top-k books similar to the query, according to cosine similarity. In LDA and LSI, we conducted many trials to find the best number of topics from 10, 50, 100 and 200 topics, and reported the best results. A Doc2vec model was created based on the book texts, with book IDs as the labels. We experimented with multiple Doc2vec parameters, including 100 or 300 dimensions and

¹¹<https://www.goodreads.com/>

¹²<https://radimrehurek.com/gensim/>

Figure 1: Recommendation accuracy of our approach against the baselines



5 or 10 window sizes, and reported the best results. Using cosine similarity, we recommend the k most similar books to the user query, which is the average of training set's book vectors. For collaborative filtering, we implemented the matrix factorization (MF) ranking method [20] in Graphlab¹³. MF represents users and items as vectors of latent factors which we chose empirically, from 8, 16, 32 and 64 factors. Given the user's binary feedback, the algorithm is fitted using logistic loss.

Parameter Settings. To search for the best parameters, we built multiple feature-based systems using a validation set of randomly selected thirty users. We experimented with 1, 5, 15, 30 and 64 neighbors in KNN, and 15 led to the best accuracy. For ET and RF, we tried 500, 1000, 1500, 2000 and 3000 trees, and 120, 60, 10, 7 and 3 features to consider at a split. It is observed that a smaller number of feature leads to better accuracy, and found that the best performance was recorded at 1000 and 3 for ET, and 1500 and 10 for RF. As it has non-competitive performance on the validation and test sets, we dropped RF from all the next results, and since many users lacked negative feedback we randomly included in the ET training set irrelevant books for the target user equal to the relevant ones. Prior to applying KNN, Z-score normalization is performed in order to transform each feature independently to a scale of zero average and one standard deviation. The system was implemented using scikit-learn¹⁴ and Graphlab. In the current experiments, the average number of books in the training set is 9.2.

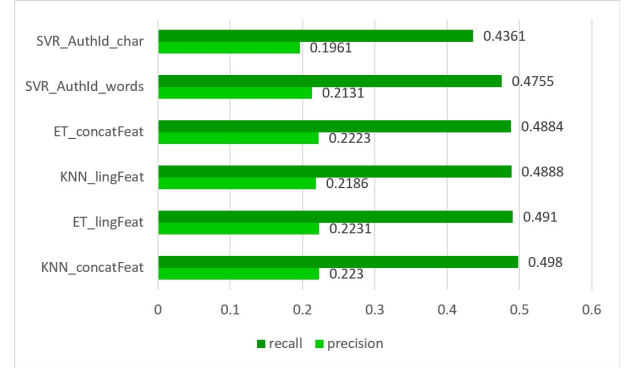
5 RESULTS AND ANALYSIS

More than a third of the relevant books in the test sets are ranked in the top ten recommended list generated by each feature-based system (Figure1). Both KNN and ET algorithms generated statistically more accurate top ten recommendation lists compared to the CB baselines, for which KNN was the most accurate. In addition, there is a statistical difference when comparing KNN-based CB to MF. All content-based baselines are significantly lower than MF, with LDA being the lowest recommendation algorithm.

¹³<https://turi.com/>

¹⁴<http://scikit-learn.org/stable/index.html>

Figure 2: Comparison of linguistic features vs. AuthID book representations from [2]



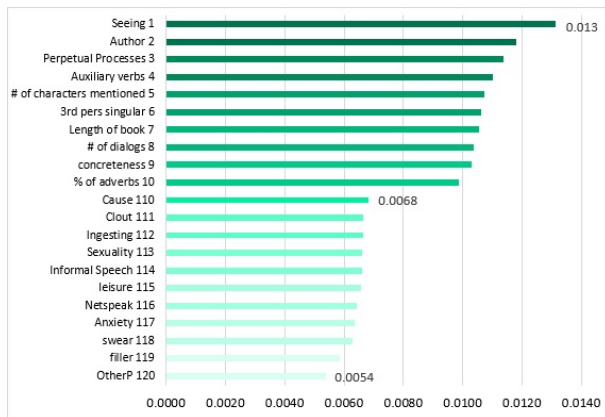
It is worth mentioning that recall@10 cannot equal one in the case of users with more than ten books in their test set, and precision@10 can only be ideal if the target user has ten or more books in their test set, which was not the case for many users in our dataset. This also explains why precision@10 is lower than recall@10.

To compare our system to [2], books in our dataset are mapped to AuthID book representations using Gutenberg ids, and users with less than 10 books are filtered out. The number of remaining users is 264 and the number of unique items is 639. We also tried concatenating our 120 linguistic features (lingFeat) with book representations (AuthId) from [2]. Two AuthID representations are tested: AuthID_words and AuthID_chars (see 2). As plotted in figure 2, both recall and precision of our two proposed approaches are higher than SVR_AuthID_words with ET_lingFeat having statistical difference in terms of recall. The use of the merged book representation has statistically increased CB and slightly decreased ET. However, AuthID book representations are based on maximumly 100,000 words and it was noticed that the more texts included in the creation of AuthID representations, the higher the system's accuracy. Since our linguistic features are based on word counting, it is not meaningful to measure frequencies of content and style words based on book excerpts (i.e., 100,000 words).

One advantage of using ET is learning what key features supported the prediction of the model. In our investigation, we considered only 730 ET models, which resulted in accurate recommendations of at least one relevant retrieved book. The variable importance values of these ET models were then averaged, and the most and least influential are presented in Figure 3. The top two variables are *Seeing*, which indicates an act of observation such as view, and *author name*. While the latter is expected to be ranked highly, the former is not. *Seeing* is strongly correlated (using Spearman) with other top-ranked features, including *concreteness* (0.59) and *perpetual processes* (0.82), which is composed of *Seeing* words plus *Hearing* and *Feeling*.

Similar to the findings by [21], the *Length of book* has a strong impact on users' reading preferences. Two characterization measurements are also important: *the number of characters mentioned* and *the number of dialogs*, both of which are positively correlated

Figure 3: Most and least influential variables in accurate ET models



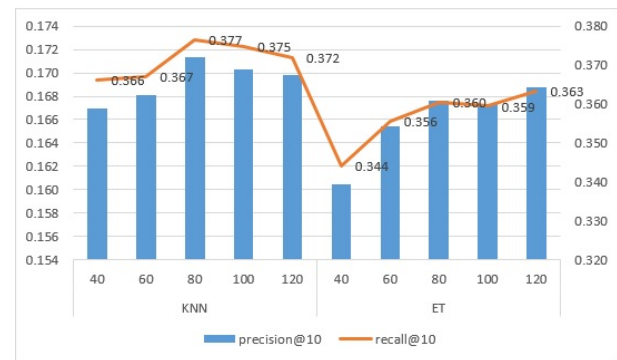
with the *length of book*. In general, all characterization and six-style features are placed above the 60th rank (i.e. the first half of the feature list). While *the percent of adverbs* is one of the top predictors, *the percent of nouns* and *adjectives* are ranked low, at 98 and 93, respectively. From the punctuation list *period* is best at the 20th rank, while the lowest is *other-punctuations* which counts only uncommon punctuations and ASCII characters. All informal language indicators, such as *Filler*, *Swear*, *Netspeak* and *Informal speech*, are ranked at the bottom. This is expected, as classical Gutenberg literary books do not typically include informal words; thus, such findings could differ in other datasets.

It would be interesting to use the previous list of top variables to generate recommendations using KNN and ET. Figure 4 shows the more features ET is fed, the higher its accuracy. This is not the case in term of KNN with increasing performance after the removal of ten and twenty least significant features.

Further analysis was conducted to determine the best and worst features for the majority of users. We counted the number of test cases (ET models) where a feature is considered the most or least influential and presented most frequently in Table 2, which overlaps the previous figure list but at different ranking. Some variables, such as *length of book*, did not appear in Table 2; though they seemed to have high significance for many users, they were not the highest. However, the top feature, *drives and needs*, is not in Figure 3, but is ranked 18th. It counts words that describe *affiliation* (e.g., ally or friend), *achievement* (e.g., winning), *power* (e.g., bully), *rewards* (e.g., prize) and *risks* (e.g., danger).

The average, maximum and minimum *drives and needs* and *seeing* values are 5.8, 12.9 and 3.42 and 1.51, 4.7 and 0.53, respectively. Of 301 relevant books for the twenty users with *drives and needs* as the best feature, only 26 have a value higher than the average, suggesting that these users tend to prefer books with less focus on *drives and needs*. Twenty users with *seeing* as the top feature have 103 relevant books out of 253, and 18 of 22 irrelevant books with *seeing* values higher than the average. It seems that these users prefer to read books with smaller *seeing* values.

Figure 4: The performance of ET and KNN when only top features are considered



A qualitative analysis was conducted by assessing if books considered similar by the KNN system share similar authors or descriptions on NoveList. We represented books as vectors of all features except for *author name*. All the top ten books similar to a randomly-selected book by Anthony Trollope were written by him as well, and eight of the top ten books by Honore de Balzac are similar to one of his books. Table 3 shows the books similar to *Madame de Treymes* by Edith Wharton. Only the first two books were written by her, though all the retrieved books have at least one shared description (highlighted in bold) with the queried book. Most have a topical resemblance, however, only two share the same storyline and character type. To determine whether characterization features have an effect on the retrieval of two books by Henry James, a reevaluation was done after removing them. One of his books (*Sir Dominick Ferrand*) was not retrieved, and one book by Emile Gaboriau, which has nothing in common with the queried book, was added.

6 CONCLUSION

This study was conducted to determine the effect of multiple linguistic and stylistic elements on reading preferences. We learn from favored books in terms of style and content to predict future preferences. The two adopted recommendation algorithms outperformed several CB and CF approaches in the top k recommendation scenario. We highlighted the variables considered important using an ensemble of randomized trees, and produced book suggestions based on them. Further qualitative analysis showed that similar books generated by our system share the same author or similar descriptions on NoveList.

Due to copyrights issues, texts of books other than Gutenberg texts were not accessible for this study. It would be interesting to conduct similar research on books from various times and different genres. Moreover, the multi-language applicability of our approach is worth investigation. Other text measurements that require extensive preprocessing, such as the number of characters in each scene, could also be applied. Also, word disambiguation could be conducted before word counting to precisely calculate word categories, which may lead to better overall accuracy.

Table 2: Most and least important features according to ET model

Top Feature	Number of test cases	Last Feature	Number of test cases
Core drives and needs, Percent of text adverbs, objective, author	25-31	Fillers	151
Seeing, Feeling	20-25	Other punctuations	45
Number of characters mentioned, Dictionary words, Period, Auxiliary verbs	15-20	Swear, Netspeak	15-25
Number of dialogs, Female referents, Discrepancies, Future focus, Abstract, Negations, Perpetual Processes, Semicolons, 3rd pers singular, Subjective, Power, Reward focus, motion, Parentheses	10-15	Friends, Sexuality, Tentativeness, Percent of characters which are female, 2nd person, Achievement, cause, home	10-15

Table 3: Top ten similar books to Madame de Treymes by Edith Wharton

Book name (similarity)	Description on NoveList
1- Madame de Treymes by Edith Wharton (1) 2- The Descent of Man and Other Stories by Edith Wharton (0.72) 3- The Glimpses of the Moon by Edith Wharton (0.71)	<i>Genre: Love stories</i> , Modern classics; <i>Character: Complex</i> ; <i>Storyline: Character-driven</i> ; <i>Writing Style: Descriptive, Lush, Richly detailed</i> ; <i>Time Period: 1920s</i> ; <i>Location: Europe – Social life and customs – 20th century</i> ; <i>Subject headings: Social status, Freeloaders, Americans in Europe, Honeymoons, Married people, Misunderstanding, Men/women relations</i>
4- Our Friend the Charlatan by George Gissing (0.71) 5- Denzil Quarrier by George Gissing (0.698)	<i>Genre: Psychological fiction, Domestic fiction, Love stories</i> , Historical fiction, Satirical fiction; <i>Time Period: 19th century</i> ; <i>Subject headings: Women – Employment - England, Middle class women - England, Women England, Single women - England, Sisters - England, Married women - England, Self-discovery in women, Authors, Fiction writing, Married people</i> ; <i>Location: London, England</i>
6- A Daughter of To-Day by Sara Jeannette Duncan (0.697)	<i>Genre: Canadian fiction</i> ; <i>Time Period: 20th century</i> ; <i>Subject headings: Brothers and sisters, Politicians, Romantic love, Clergy, Political science, Men/women relations</i> ; <i>Location: Ontario</i>
7- In the Year of Jubilee by Gissing (0.69)	See above
8- The Tragic Muse by Henry James (0.687) 9- Sir Dominick Ferrand by Henry James (0.682)	<i>Genre: Classics, Psychological fiction</i> ; <i>Character: Complex, Flawed</i> ; <i>Storyline: Character-driven</i> ; <i>Pace: Leisurely paced</i> ; <i>Tone: Melancholy, Reflective, Thought-provoking</i> ; <i>Writing Style: Stylistically complex</i>
10- The Tragic Bride by Francis Brett Young (0.678)	<i>Genre: Domestic fiction</i> <i>Subject headings: Young women, Quests, Love, Men/women relations, Marriage, Remarriage, Self-discovery in women, Happiness in women</i> ; <i>Location: Midlands, England</i>

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